

RESEARCH AGENT

AI-Powered Research Assistant

Problem Statement

In today's digital environment, researchers, students, and professionals have access to a vast amount of information from websites, academic papers, documents, and other online sources. However, finding relevant information, verifying sources, organizing research materials, summarizing large amounts of content, and converting findings into useful reports can be time-consuming and difficult. Traditional search tools mainly provide links or isolated pieces of information and often require users to manually read, compare, organize, and analyze the collected material.

There is therefore a need for an intelligent and user-friendly platform that can assist users throughout the research process. The proposed **Research Agent** aims to address this problem by using Artificial Intelligence and Generative AI to understand research queries, discover and organize relevant information, summarize content, assist with analysis, manage research projects, and generate structured research outputs.

Proposed Solution

Research Agent is a web-based AI-powered research assistant designed to simplify and accelerate the research workflow. Users can enter a research topic or question, collect relevant sources, interact with an AI assistant, organize information into projects, and generate concise research reports. The system will provide a centralized workspace where research activities can be managed efficiently.

Key Objectives

- Provide an intelligent interface for entering and exploring research questions.
- Use AI to summarize and organize information from relevant sources.
- Help users manage research topics, sources, notes, and projects in one place.
- Reduce the time required to manually search, read, and organize large amounts of information.
- Generate structured research reports and useful insights from collected information.
- Provide a simple and accessible interface suitable for students, researchers, and professionals.

Major Features

Feature	Description
AI Research Assistant	Answers research-related queries and assists users during research.
Research Workspace	Provides a centralized space to manage an ongoing research task.
Source Management	Allows users to collect, organize, and review research sources.
AI Summarization	Produces concise summaries from lengthy research material.
Project Management	Enables users to save and organize multiple research projects.

Feature	Description
Report Generation	Creates structured outputs based on collected research information.
Dashboard & Analytics	Displays research activity, projects, and useful visual insights.

Project Domain

Research & Knowledge Management

Core technologies/domains: Artificial Intelligence, Generative AI, Natural Language Processing (NLP), Information Retrieval, Data Analytics, and Full-Stack Web Development.

Expected Outcome

The proposed system is expected to provide a centralized, intelligent, and efficient research environment that reduces manual effort and helps users move from a research question to organized information and structured insights more quickly. The system should also improve research organization and make AI-assisted research accessible through an easy-to-use web interface.

Target Users

- Students working on academic projects and assignments.
- Researchers conducting literature reviews and information gathering.
- Professionals preparing reports, market research, or technical studies.
- Educators and learners who need organized research information.